ModelIt!

Student Activity Pack

Urban Heat Islands: Why Cities Are Hotter Than Forests

Modeling How Built Environments Amplify Extreme Heat

Grade 09L2 | NGSS HS-ESS3-4, HS-PS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think cities are hotter than surrounding countryside?

2 Have you noticed temperature differences between different neighborhoods? Why might that be?

3 Draw a diagram showing how concrete, trees, and air conditioning might affect local temperature.

4 What is environmental justice and how might it relate to heat?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Impervious Surface Area | Green Space Coverage | Ambient Temperature | Energy Consumption (AC) | Air Quality Index | Heat-Related Health Risk

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Neighborhood Comparison

Wealthy area vs. underserved area

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heat Wave Amplification

Baseline +5 C with urban heat island

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Green Intervention

Green space from 10 to 40 percent

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cool Roof Coverage — Percentage of buildings with reflective roofing that reduces heat absorption

Public Transit Usage — Percentage of commuters using public transit vs. cars, reducing vehicle waste heat

Nighttime Temperature — Minimum temperature at night, which fails to drop in dense urban areas due to stored heat release

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why can a city be 10 degrees hotter than the forest just 20 miles away, and who suffers most?
Use evidence from your simulation to support your answers.

1 How does your model show that the urban heat island is a self-reinforcing system?

2 Why do wealthy and poor neighborhoods experience such different temperatures?

3 How does air conditioning act as both a solution and a cause of urban heat?

4 What does environmental justice mean in the context of heat vulnerability?

5 What trade-offs exist between rapid cooling solutions and long-term structural change?

8 STEM Challenge: Design an Equitable Urban Cooling Plan

ModelIt!

YOUR ENGINEERING MISSION

Design a neighborhood cooling strategy that prioritizes environmental justice by targeting interventions to the most heat-vulnerable communities.

THE SCENARIO

A city has funding for urban cooling but limited budget. Some neighborhoods are 10+ degrees hotter than others. Design a plan that allocates cooling interventions equitably based on heat vulnerability, not political influence.

Engineering Design Thinking

1. HOW WOULD YOU MAP HEAT VULNERABILITY ACROSS THE CITY?
2. WHAT COMBINATION OF GREEN INFRASTRUCTURE AND COOL SURFACES WOULD YOU DEPLOY?
3. HOW WOULD YOU ENSURE THAT COOLING INVESTMENTS DO NOT CAUSE GENTRIFICATION?
4. HOW WOULD YOU MEASURE WHETHER YOUR PLAN ACTUALLY REDUCED HEAT INEQUALITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Urban Climate Scientists study how built environments modify local weather and design heat mitigation strategies, earning $60,000-$120,000/year at city planning departments, universities, and environmental

firms.